

humanovo

*A search engine for the **unwritten** — turning the published biomedical literature into testable, mechanism-aware hypotheses, in minutes instead of weeks.*

FOUNDER

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Founder & Engineer

STAGE

Pre-seed
Live product

SECTOR

AI · Scientific tools
B2B SaaS

RAISING

\$1,000,000
Speedrun SAFE

The literature already *contains the answers.*
Nobody can read *five million papers a year.*

5*M*+

Academic & scientific papers
published globally each year, 2026.

13*K*/day

Roughly 5,500–13,000 new papers
published every single day.

~90%

Of clinical trials fail — most because
the underlying hypothesis was wrong.
*Often, contradicting evidence existed in
the literature before the trial began.*

Sources: WordsRated / Scopus / Web of Science publication statistics 2025–2026; BIO / Biomedtracker on Phase I–III clinical-trial attrition.

Three forces, in *sequence*, in the last eighteen months.

I. MODELS

Reason over full-text.

Frontier LLMs can finally hold an entire paper in context — methods, figures, supplementary. The unit of synthesis changed from abstracts to whole arguments.

II. ACCESS

Corpus is addressable.

Open-access mandates pushed millions of papers outside paywalls. For the first time, you can build a real product on top of the literature — with licensed full-text access on the roadmap.

III. METHOD

Adversarial grounding.

Multi-model adversarial pipelines catch the hallucinations that single-model RAG misses. Trustworthy generation finally became possible.

Plain English in. *Cited hypotheses* out.

A researcher types a hypothesis question in plain English. humanovo returns a ranked list of testable hypotheses — each grounded in cited papers, mechanism made explicit, every claim auditable back to source.

INPUTS

Natural-language questions across *treatment, prevention, biomarker, repurposing*, combination.

PIPELINE

A *twelve-stage adversarial* chain of frontier models, with dual-embedding grounding between every stage.

OUTPUTS

Ranked hypotheses with *full provenance* — sentence-level citations, mechanistic chain, translational roadmap.

EXAMPLE

“What mechanisms link GLP-1 agonists to neuroprotection in Parkinson's?” → *4 ranked hypotheses*, 23 cited papers, 3 translational pathways — generated in 4 minutes.

Live product. Demo available on request.

Twelve adversarial stages. Every claim *grounded twice*.

- 01–02 *Seed & Expand*. Claude Opus generates the initial hypothesis; Sonnet broadens scope.
- 03–05 *Evidence, Counter, Revise*. Cohere reviews the literature; Mistral attacks the hypothesis; o3-mini revises.
- 06–08 *Mechanism, Validate, Ground*. GPT-4.1 builds the mechanism; Sonnet cross-validates; Grok-4 grounds in three layers.
- 09–10 *Score & Refine*. Multi-dimensional confidence scoring, then fast refinement.
- 11–12 *Translate & Finalize*. Translational roadmap T0–T5; final synthesis with full citations.

BETWEEN EVERY STAGE

Dual-embedding grounding — two embedding models (1024d and 1536d) running in parallel.

Every claim is RAG-retrieved against *60+ data sources* — PubMed, ClinicalTrials.gov, openFDA, UniProt, Reactome, ChEMBL, AlphaFold — and ungrounded claims are gated out before the next stage runs.

*A hypothesis that takes a postdoc **weeks** to assemble — sifting, reading, cross-referencing, arguing with their PI — is the unit of cost we are trying to compress.*

I .

Adversarial generation, not retrieval, is the unlock for **trustworthy** AI in science.

II .

Mid-tier institutions are where bottlenecked human capital meets **willingness to pay**.

III .

A platform for **hypotheses** compounds in a way a platform for papers cannot.

In active use across *four UPMC departments*:
Radiation Oncology, Neuroimaging, Gastroenterology, Psychiatry.

“It does a *fantastic job* of summarizing potential hypotheses and proposing possible methods to solve it.”

— UPMC RESEARCHER
Unprompted, in-product feedback

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RECOGNITION

NYBPC Concept Award
Winner

FOUNDER BACKGROUND

B.Tech Biotechnology · Research at *Glenmark Pharmaceuticals* — where this problem became obvious.

B2B from day one.

Free trial, then paid.

RESEARCHER

\$20 / mo · per seat

Individual scientists, postdocs, students

Limited monthly hypotheses
Caps designed for light use

Core discovery modes & citation export

Top-ups available
Buy extra usage as needed

LAB

\$200 / mo · per lab

Active labs, PIs, biotech R&D teams

Generous hypothesis volume
For sustained research workflows

All discovery modes, API access, projects

Burst top-ups for grant cycles

INSTITUTION

Custom

Universities, pharma, research consortia

Dedicated infrastructure & SLAs

Private corpus integration, SSO, audit logs

Solutions engineering & onboarding

TRIAL 3 hypotheses + 1 research-paper synthesis, free, before signup converts.

Search. Summarize.
Adversarially generate.

CAPABILITY	PUBMED	CHATGPT	ELICIT	CO-SCIENTIST	HUMANOVO
Reasons over full-text papers	—	≈	✓	✓	✓
Adversarial multi-model pipeline	—	—	—	≈	✓
Dual-embedding claim grounding	—	—	—	—	✓
Generates novel hypotheses	—	≈	—	✓	✓
Auditable, sentence-level citations	✓	—	✓	≈	✓
Available to working PIs today	✓	✓	✓	—	✓

✓ meets ≈ partial — absent

Google AI Co-Scientist is closed beta — restricted to selected Google partner institutions.
humanovo is the first adversarial hypothesis engine available to working PIs today.

Where the *willingness to pay* actually lives.

BEACHHEAD

~12*K*

INSTITUTIONS

~\$8.6M ARR ceiling
12K × ~3 seats × \$20/mo × 12mo

U.S. mid-tier academic medical centers & research universities — bottlenecked PIs, real budgets, no Big-Pharma AI internal stack.

ADJACENT

~6*K*

TEAMS

~\$14M ARR ceiling
6K × \$200/mo lab tier × 12mo

Biotech R&D teams & CROs running early-stage discovery — pricing aligned with seat-based research-tool spend.

HORIZON

Pharma

TOP-20

\$20M+ contract ceiling
\$1M+ per six-figure+ contract

Top-20 pharma — discovery, repurposing, biomarker programs. Custom contracts entered via institutional pull, not cold sales.

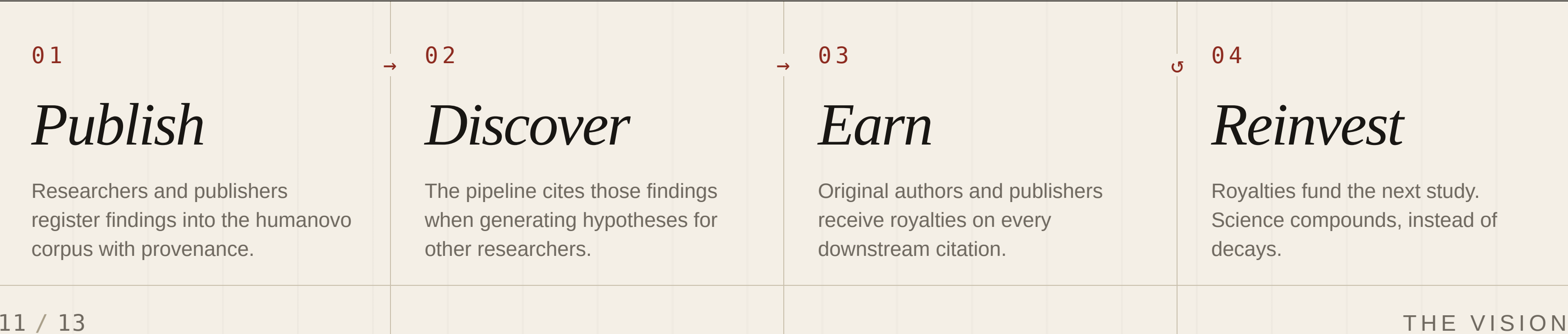
We are pricing against *a postdoc's time* — and labs already pay \$60k+/yr for that.

Counts are public NIH / Carnegie / NSF institution rolls; see also AAMC member institutions and BIO biotech-employer reports. Pricing reflects published seat-based research-tool comps.

11 / 13 THE VISION
PHASE TWO — after the engine becomes infrastructure

A journal that *pays* the people it stands on.

PHASE ONE the hypothesis engine. PHASE TWO humanovo becomes a *repository*. Researchers and publishers register their work; when their hypotheses or findings are cited inside other humanovo-derived work, they earn royalties on the downstream value. Discovery, attribution, and compensation finally close the loop.



\$1M

Twelve months to land design partners, harden the pipeline, and convert active UPMC usage into signed pilots — on a Speedrun standard SAFE.

USE OF FUNDS

~55%	Engineering	.
	Senior hires for the pipeline & grounding layer	
~25%	Compute & data	.
	Frontier-model inference, full-text licensing	
~15%	GTM	.
	Design-partner motion, conferences, content	
~5%	Reserve	.
	Legal, compliance, runway buffer	

IN CLOSING

*Every breakthrough treatment starts with a hypothesis.
We are building the place where the **next ten thousand** of them are born.*



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Founder · humanovo

humanovo.net